2022

COMPUTER SCIENCE AND ENGINEERING

Paper : CSCL-1002

(Topics in DBMS)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question nos.*** **1, 2** and ***any four*** questions from the rest.

1. Answer the following (***any five***) : 2×5

(a) What is a derived horizontal fragmentation? Give an example.

(b) Define image of a database in a site for spatially distributed database system.

(c) What is join graph?

(d) What is the role of name nodes in HDFS?

(e) Why is concept hierarchy important in designing a data warehouse?

(f) What is the purpose of Pivot operation on a cube data model?

2. Justify or disprove the following comments (***any five***) : 4×5

(a) A distributed database is not necessarily a spatial database.

(b) Dicing does not necessarily reduce dimensions of a data cube.

(c) Relational data model may not be very efficient for a semi-structured database application.

(d) Horizontal fragments are necessarily disjoint.

(e) Client application invoked from computers at different locations, all connected to a remote database server with a back-up server, is an example of a distributed database.

(f) Allocation schema is location independent although it defines the allocation of fragments for a distributed database system.

3. (a) Define min-term predicate.

(b) Briefly describe the methodology for identification of primary horizontal fragments towards the design of a distributed database system.

(c) In the context of distributed database design, why is it important to identify which part of application data are read-only items? 2+6+2

4. (a) Explain with an example how a single updation operation on a single data item could result in multiple updation operations on several sites when the database is physically distributed.

(b) In the context of distributed database design, why is it important to identify which part of application data are read-only items?

(c) Compare the advantages and disadvantages of centralized versus distributed database systems. 4+3+3

5. (a) Explain the roles of Distributed Transaction Manager (DTM) and Local Transaction Manager (LTM) towards transaction management.

(b) A 'commit' operation in database systems ensures that the changes in the database are permanent. In the 2-phase commit protocol for transaction management, two different types of 'commit' operations are performed at the coordinating site and at the participating sites. The coordinating site may decide to abort a transaction even if one or more participating site is ready for 'local commit'. Explain if this is consistent with the ACID properties for transaction management.

(c) How is the atomicity of a transaction ensured under 2-phase commit protocol when the coordinator site fails just before sending out a global COMMIT or ABORT message? 4+3+3

6. (a) ITRA funded remote healthcare project CUDFIKJ requires to store data pertaining to different patients, their ailment, doctors and other care givers. A patient may need advice from one or more doctor(s), depending on the disease and availability of resources. The discomforts or symptoms recorded for patients vary a lot from case to case.

We assume that a Hyper file named *PatientComplaints* stores the basic information like patient name, address, age followed by one or more discomforts. The parameters for discomfort change on its nature. As for example, if a patient complains about pain, information is recorded on the organ where the pain is felt, nature of pain (continual or periodic, etc.), etc. On the other hand, if the patient complains about fever, its important to store the body temperature.

The employee profile too varies depending on the types of employees and their job functionality. We assume that a Hyper file named *HealthServices* stores the basic information like service-title, doctor-id, major-symptom, and cost of different services. Assume that some of the entries in the Hyper file *HealthServices* point at some other entries as FRT (frequently required together), if such services are often requested together.

(i) Write a query using Hyper file model to list the name and age for all the Patients treated in the CUDFIKJ project.

(ii) Write a query using Hyper file model to extract FRT services for patients coming with chest pain complaints as major symptom. Your query must extract all the items that are linked up to third level considering chest-pain in level 1.

[State your assumptions clearly for the additional Hyper files, if any, in the database other than the two mentioned above that you may need to define for answering the queries.] 4+6

7. (a) A cube model has data in four dimensions for a particular subject orientation. The cardinalities for the four dimensions are 3, 4, 5 and 4 respectively.

   (i) How many Fact and Dimension Tables are required for an equivalent Star schema?

   (ii) State and explain the minimum number of tuples that would occur in each of the Dimension and Fact Tables in your design.

   (b) How do you relate lattice structure with cube-data model? Explain with a small example. 6+4

8. (a) Describe the 6-layer architecture for a distributed database system and state the roles for each layer.

   (b) What is semi-join? How could semi-join operation significantly reduce the cost of query processing in a distributed environment? 6+4